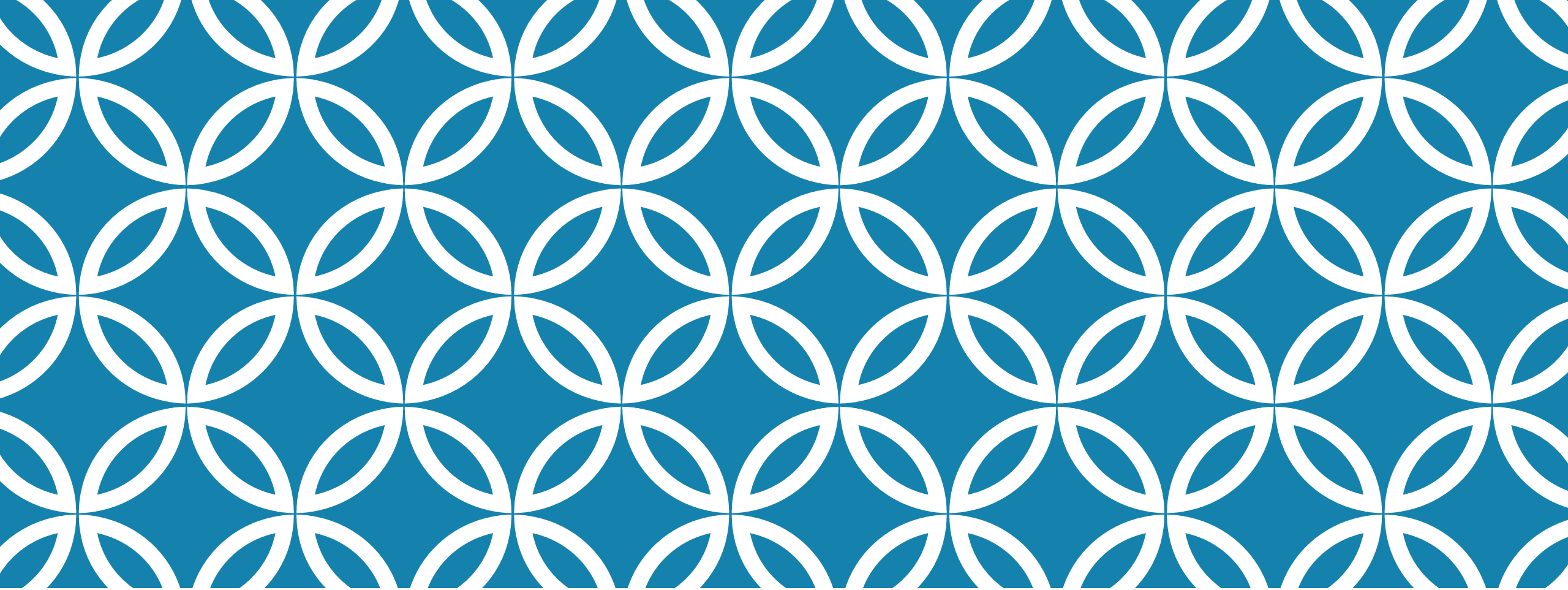


LALR PARSER

LALR (1) PARSER

LALR stands **for Look Ahead LR parser**

It uses **LR(1)** canonical items

Size of the **LALR parsing table is small** as compared to CLR parser

The 1 in LALR(1) is a lookahead item or symbol

LR(1) ➔ LR(0) + lookahead

In LALR parsing table, we will **combine states** that have **similar set of productions** but their **lookahead is different.**

Hence, the states as compared to CLR parser table will be less

EXAMPLE – 1

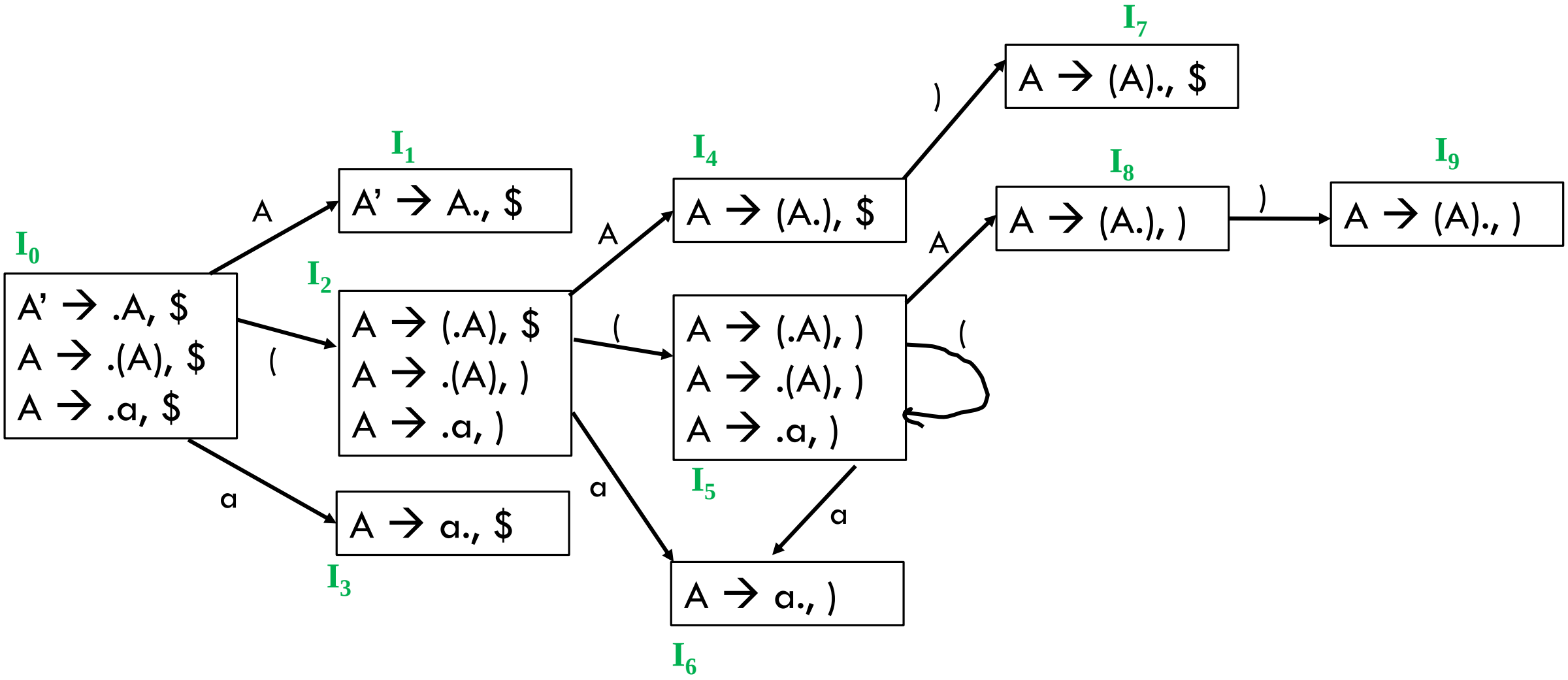
A → (A) | a

Let us find out whether this grammar can be parsed using LALR parser

0: A' → .A

1: A → .(A)

2: A → .a


I0
A' → .A, $
A → .(A), $
A → .a, $
A
(
a
I1
A' → A., $
I2
A → (.A), $
A → .(A),)
A → .a,)
A
(
a
I3
A → a., $
I4
A → (A.), $
I5
A → (.A),)
A → .(A),)
A → .a,)
A
(
a
I6
A → a.,)
)
I7
A → (A)., $
I8
A → (A.),)
)
I9
A → (A).,)

LALR(1) Parser table

States	Action				Goto
	(	)	a	$	A
I_0	S_{25}		S_{36}		1
I_1				Accept	
I_{25}	S_{25}		S_{36}		48
I_{36}		r_2		r_2	
I_{48}		S_{79}			
I_{79}		r_1		r_1	

Since no shift-reduce or reduce-reduce conflict, grammar can be parsed using LALR parser

EXAMPLE – 2

S → L = R | R

L = *R | id

R → L

Find out whether this grammar can be parsed using LALR parser

LALR(1) Parser table

States	Action				Goto		
	=	*	id	$	S	L	R
I_0		S411	S512		1	2	3
I_1				Accept			
I_2	S6			R5			
I_3				R2			
I_{411}		S411	S512			810	713
I_{512}	R4			R4			
I_6		S411	S512			810	9
I_{713}	R3			R3			
I_{810}	R5			R5			
I_9				R1			

EXAMPLE – 3

S → Aa | bAc | Bc | bBa

A → d

B → d

Find out whether this grammar can be parsed using LALR parser

LALR(1) Parser table

States	Action					Goto		
	a	b	c	d	$	S	A	B

Try out exercises from other parsers